

EXPERIMENT

STS30 TEMPERATURE SENSOR INTERFACING WITH DEV BOARD/NODE

What will you learn from this module:

- Interfacing with the help of I2C protocol.
- Temperature measurement using STS30 sensor using nrf dev board/node.
- Configuration of overlay file, device tree and prj file for enabling hardware device.

Requirements:

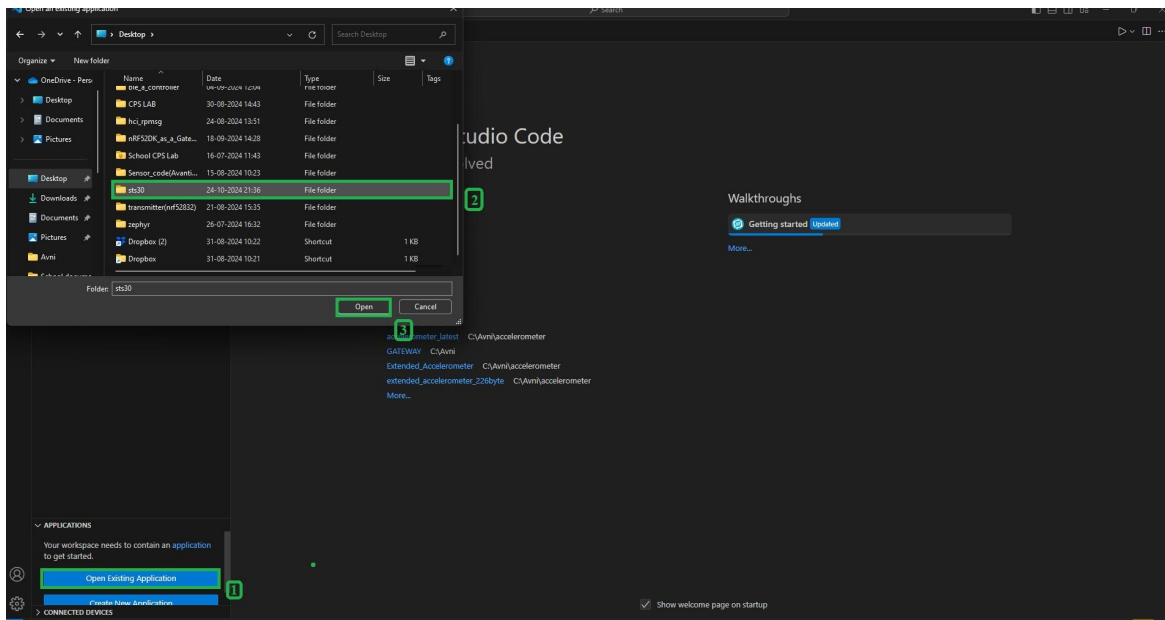
- nRF connect desktop software.
- nRF Command line tools.
- Visual studio code.
- USB cable.
- nRF52832 Development Board/Node.
- STS30 Sensor.
- TTL Device.

Prerequisites:

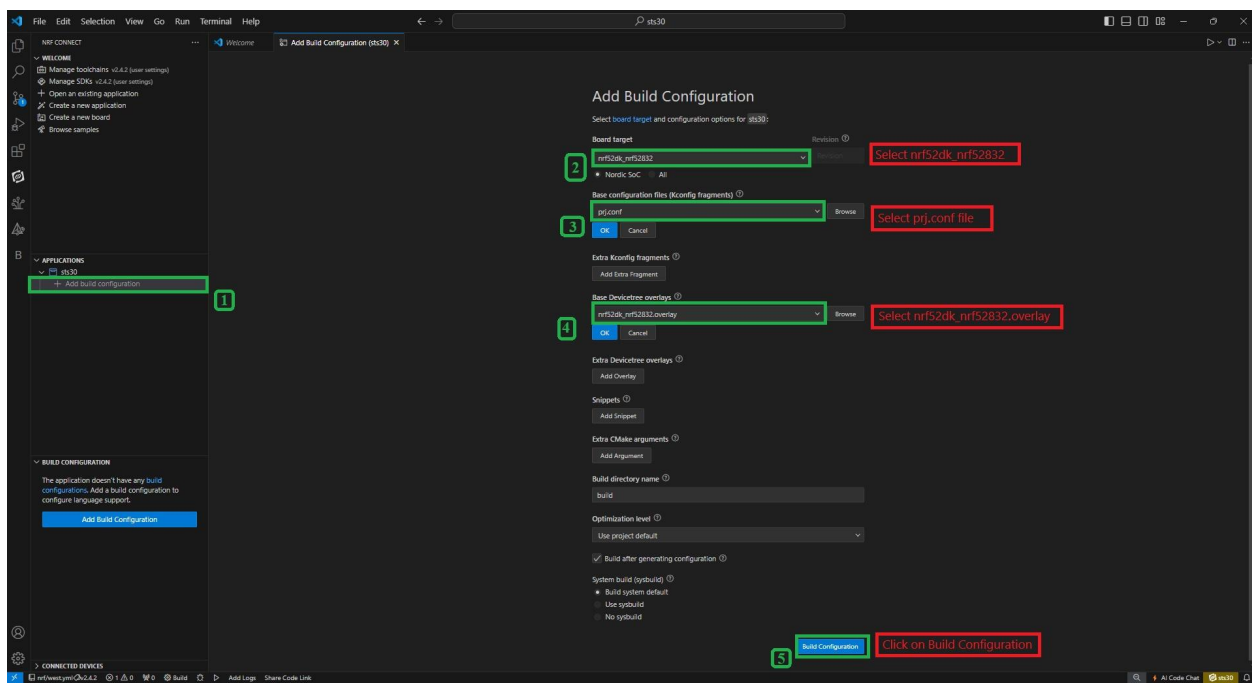
- Basic knowledge of C/C++.
- Basic knowledge of communication protocol.
- Basic project setup.

Setup and Configuration:

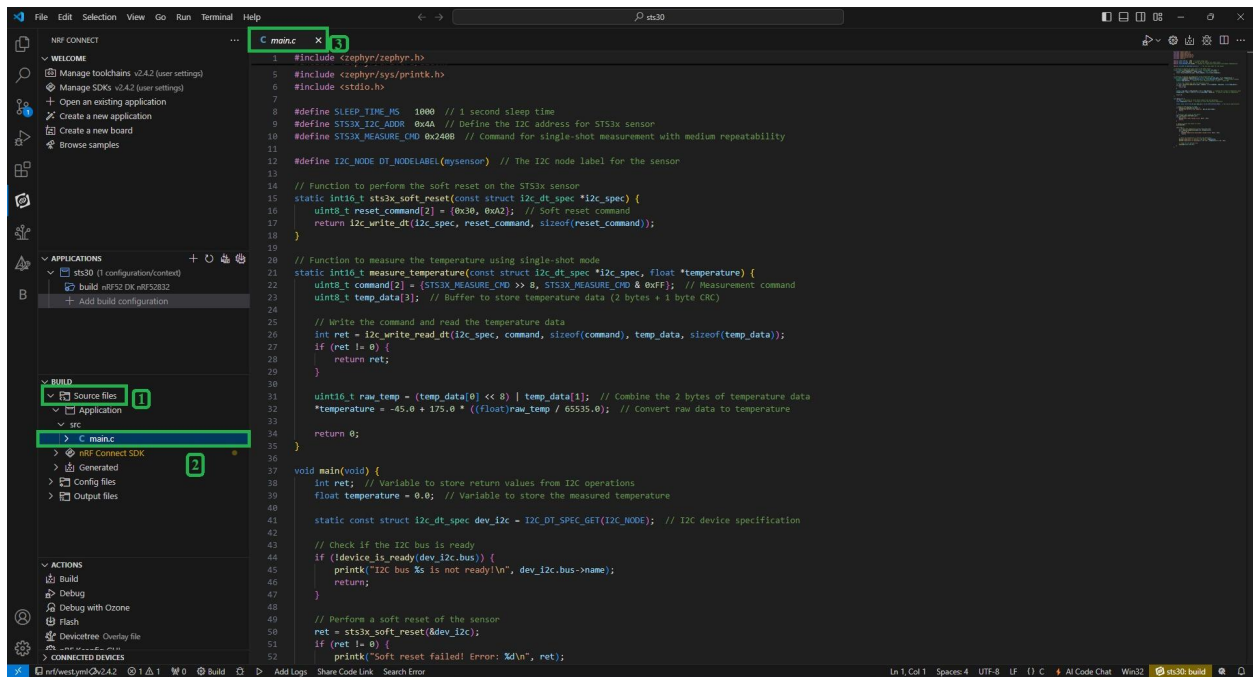
- Open VS Code and click on **Open Existing Application [1]** > click on **STS30 [2]** > **Open [3]** as shown in the picture below.



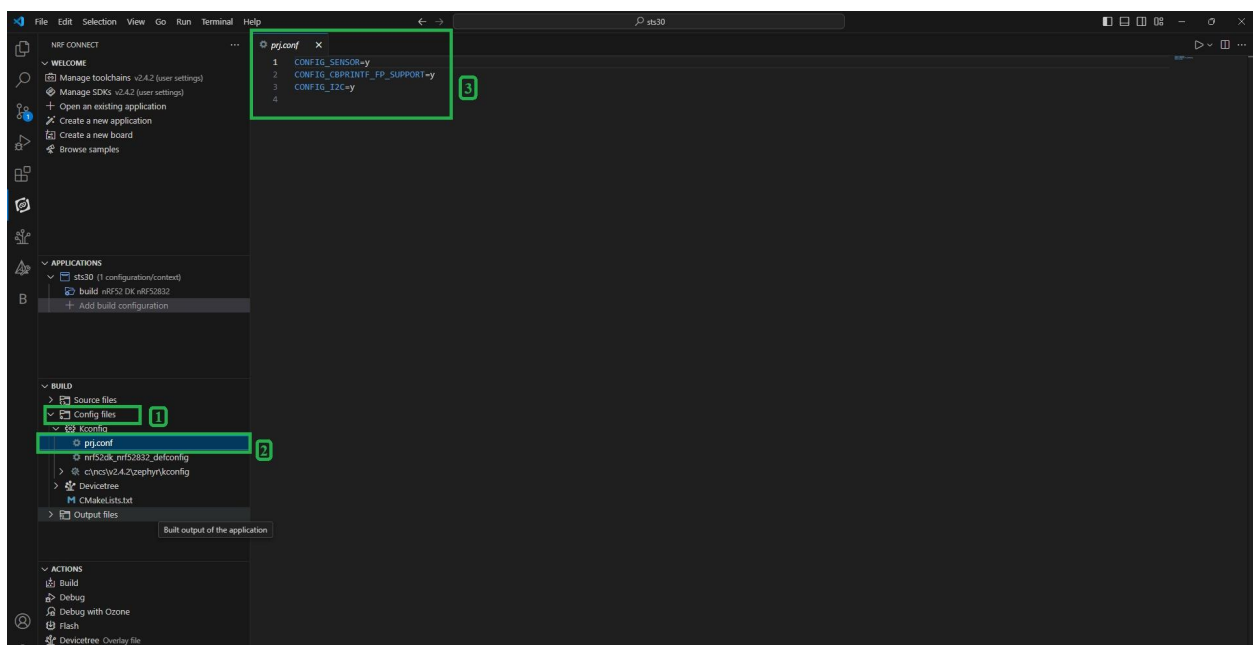
- Click on **Create new build configuration [1]**. Here you can change the board version, if you are using nRF52832, then select **nrf52dk_nrf52832 [2]** or you can change from dropdown menu for another version like nRF52833 etc.
- Click on the Configuration and select **prj.conf [3]** from dropdown menu and then click on the device tree overlays and select **nrf52dk_nrf52832.overlay[4]**.
- Then click on the **Build Configuration [5]** as shown below in the picture.



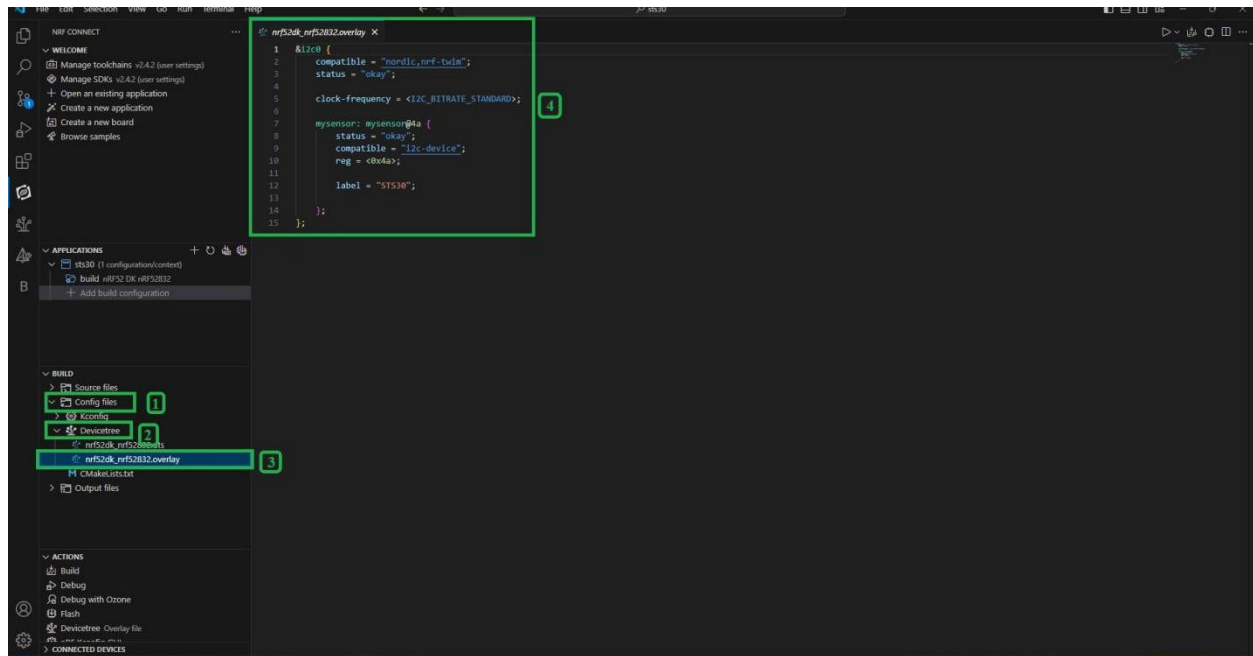
- Go to source file, click **source file [1]** > click on **Application** > click on **src** > click on **main.c [2]**.
- By Clicking on **main.c** file and you will see the code on your screen [3].



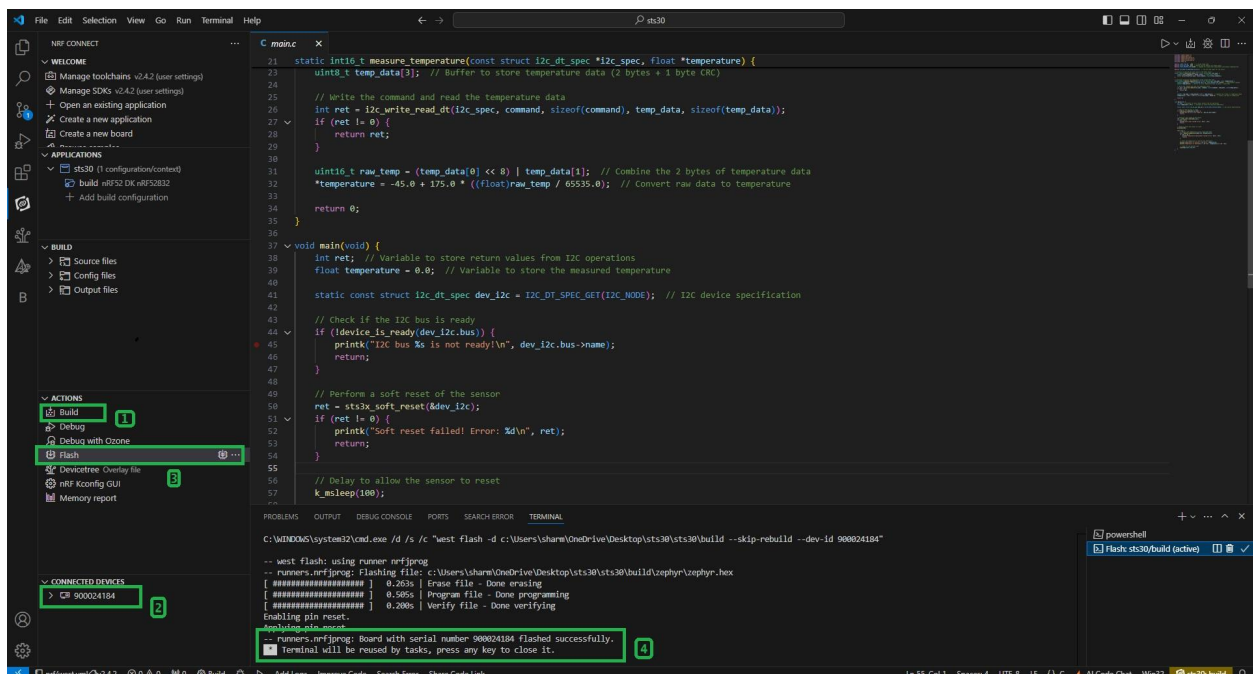
- To configure the prj configuration, click on **Config files [1]** > click on **Kconfig** > click on **prj.conf [2]**.
- The prj configuration will appear on your screen [3] as shown in the picture below.



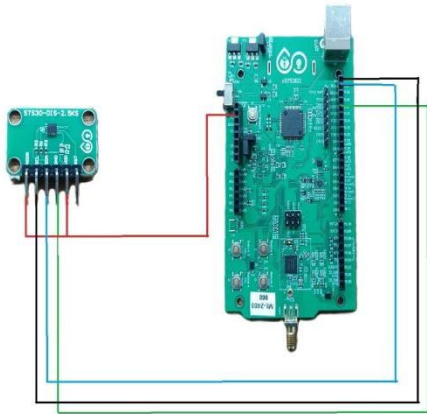
- To configure the i2c protocol, you have to enable it in the **.overlay** file.
- Click on the **Config files [1]** > click on **Kconfig** > click on **Devicetree [2]** > click on **nrf52dk_nrf52832.overlay [3]**.
- The .overlay file will appear on your screen and add the given code to the .overlay file as shown in the picture given below [4].



- Click on **Build [1]** configuration again and check the **CONNECTED DEVICES [2]**.
- If device id is visible, then **Flash [3]** the code in Dev Kit.
- If **flashed successfully [4]** message is displayed on serial terminal, then flash process is complete.



❖ PIN CONFIGURATION



Board Pins -> Sensor Pins

GND -> GND

VDD(3.3V) -> VDD

PO.26 -> SDA

PO.27 -> SCL

VDD -> ADDR

❖ OUTPUT

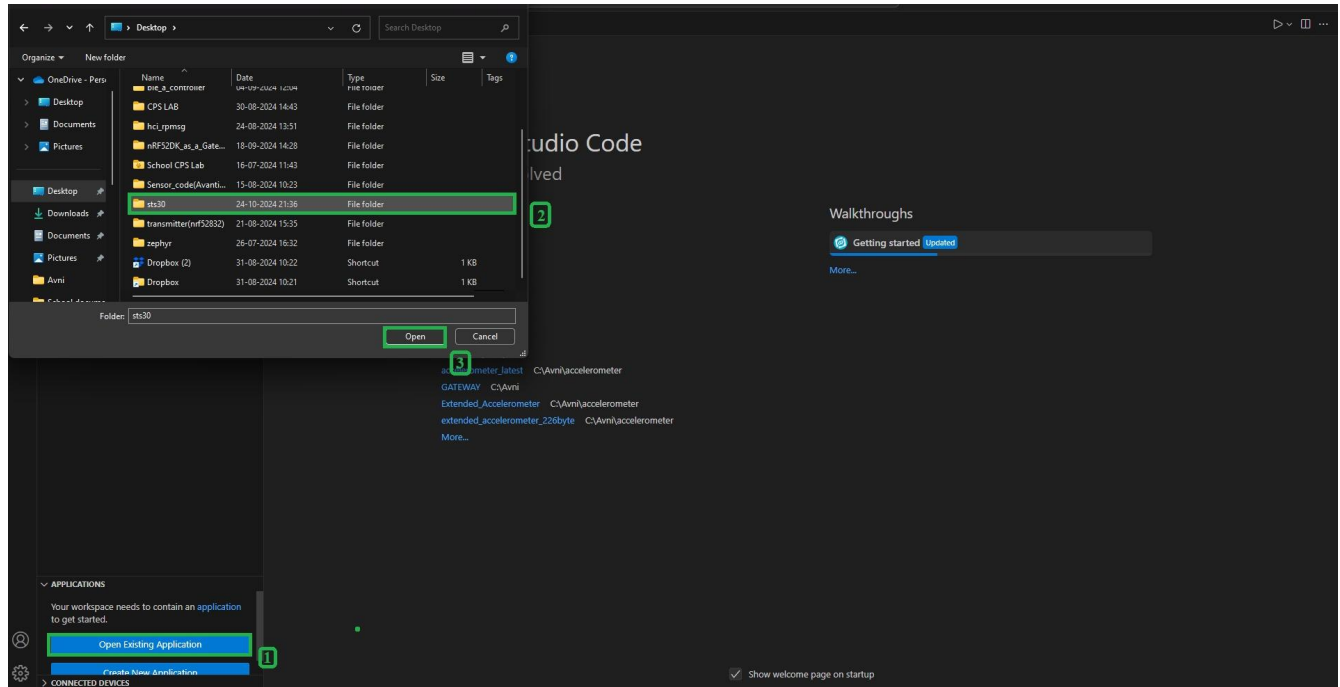
```
src > C main.c ...
1 #include <zephyr/zephyr.h>
2 #include <zephyr/device.h>
3 #include <zephyr/device_tree.h>
4 #include <zephyr/drivers/i2c.h>
5 #include <zephyr/sys/printk.h>
6 #include <stdio.h>
7
8 #define SLEEP_TIME_MS 1000 // 1 second sleep time
9 #define STS3X_I2C_ADDR 0x4A // Define the I2C address for STS3X sensor
10 #define STS3X_MEASURE_CMD 0x2000 // Command for single-shot measurement with medium repeatability
11
12 #define I2C_NODE DT_NODELABEL(i2c0) // The I2C node label for the sensor
13
14 // Function to perform the soft reset on the STS3X sensor
15 static int i2c_soft_reset(const struct i2c_dev *i2c_dev) {
16     uint8_t reset_command[2] = {0x80, 0xA2}; // Soft reset command
17     return i2c_write_dt(i2c_dev, reset_command, sizeof(reset_command));
18 }
19
20 // Function to measure the temperature using single-shot mode
21 static int i2c_measure_temperature(const struct i2c_dev *i2c_dev, float *temperature) {
22     uint8_t command[2] = {STS3X_MEASURE_CMD >> 8, STS3X_MEASURE_CMD & 0xFF}; // Measurement command
23     uint8_t temp_data[3]; // Buffer to store temperature data (2 bytes + 1 byte CRC)
24
25     // Write the command and read the temperature data
26     int ret = i2c_write_read_dt(i2c_dev, command, sizeof(command), temp_data, sizeof(temp_data));
27     if (ret != 0) {
28         return ret;
29     }
30
31     uint16_t raw_temp = (temp_data[0] << 8) | temp_data[1]; // Combine the 2 bytes of temperature data
32     *temperature = -45.0 + 175.0 * ((float)raw_temp / 65535.0); // Convert raw data to temperature
33
34     return 0;
35 }
36
37 int main(void) {
38     struct i2c_dev *i2c_dev;
39     float temperature;
40
41     // Get the I2C device pointer
42     i2c_dev = device_get_ptr(i2c0);
43
44     // Perform soft reset
45     i2c_soft_reset(i2c_dev);
46
47     // Measure temperature
48     i2c_measure_temperature(i2c_dev, &temperature);
49
50     // Print temperature
51     printk("Temperature in Celsius: %f\n", temperature);
52     printk("Temperature in Fahrenheit: %f\n", temperature * 9/5 + 32);
53
54     // Sleep for 1 second
55     msleep(SLEEP_TIME_MS);
56
57     // Repeat the measurement
58     i2c_measure_temperature(i2c_dev, &temperature);
59     printk("Temperature in Celsius: %f\n", temperature);
60     printk("Temperature in Fahrenheit: %f\n", temperature * 9/5 + 32);
61
62     return 0;
63 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR

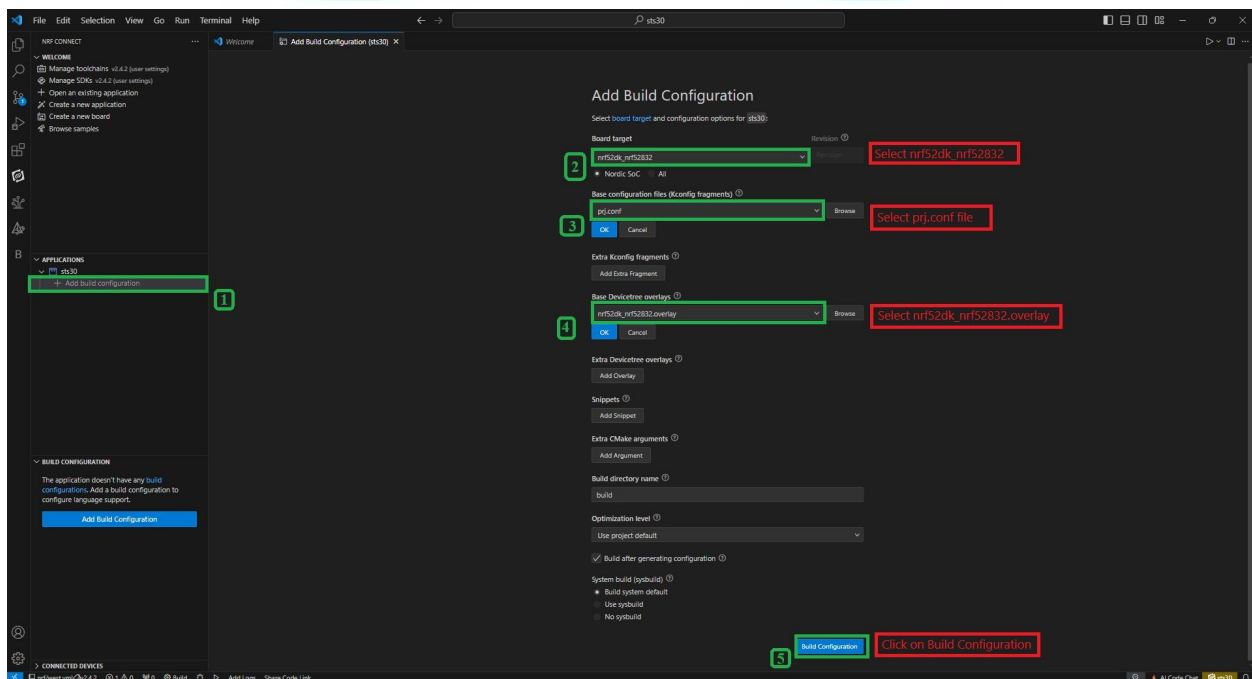
Temperature in Celsius: 26.78 °C
Temperature in Fahrenheit: 80.21 F
Temperature in Celsius: 26.77 °C
Temperature in Fahrenheit: 80.19 F
Temperature in Celsius: 26.74 °C
Temperature in Fahrenheit: 80.14 F
Temperature in Celsius: 26.76 °C
Temperature in Fahrenheit: 80.16 F
Temperature in Celsius: 26.81 °C
Temperature in Fahrenheit: 80.44 F
Temperature in Celsius: 26.83 °C
Temperature in Fahrenheit: 80.29 F

INTERFACING WITH THE HELP OF NODE

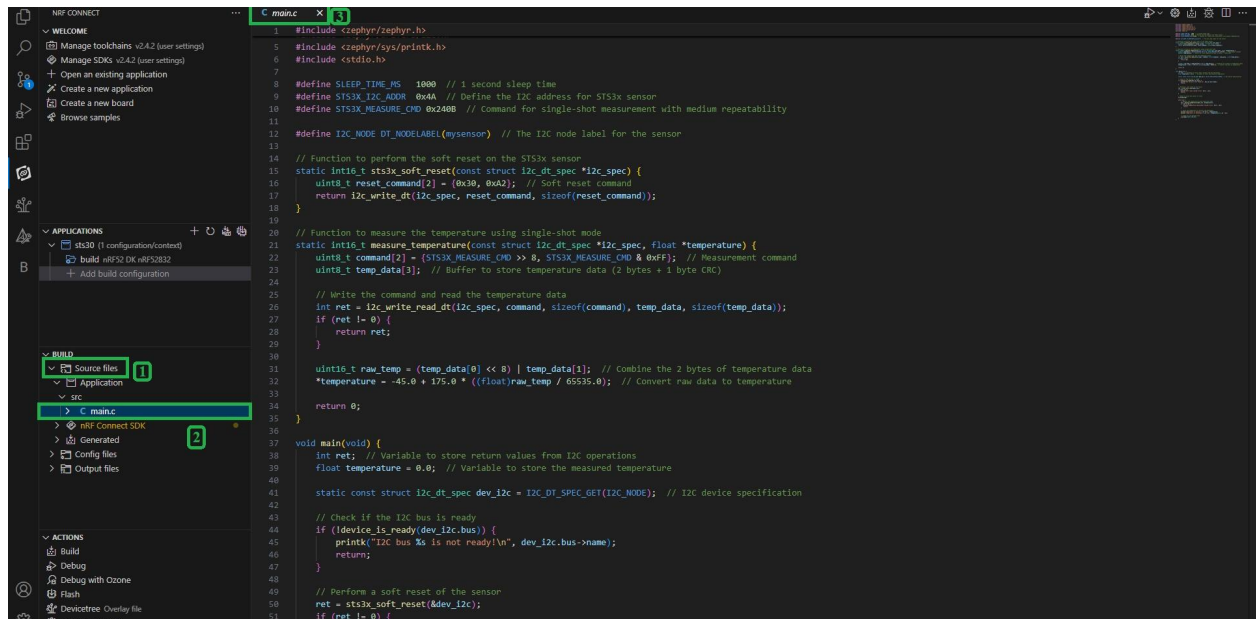
- Open VS Code and click on **Open Existing Application [1]** > click on **STS30_node_code [2]** > **Open [3]** as shown in the picture below.



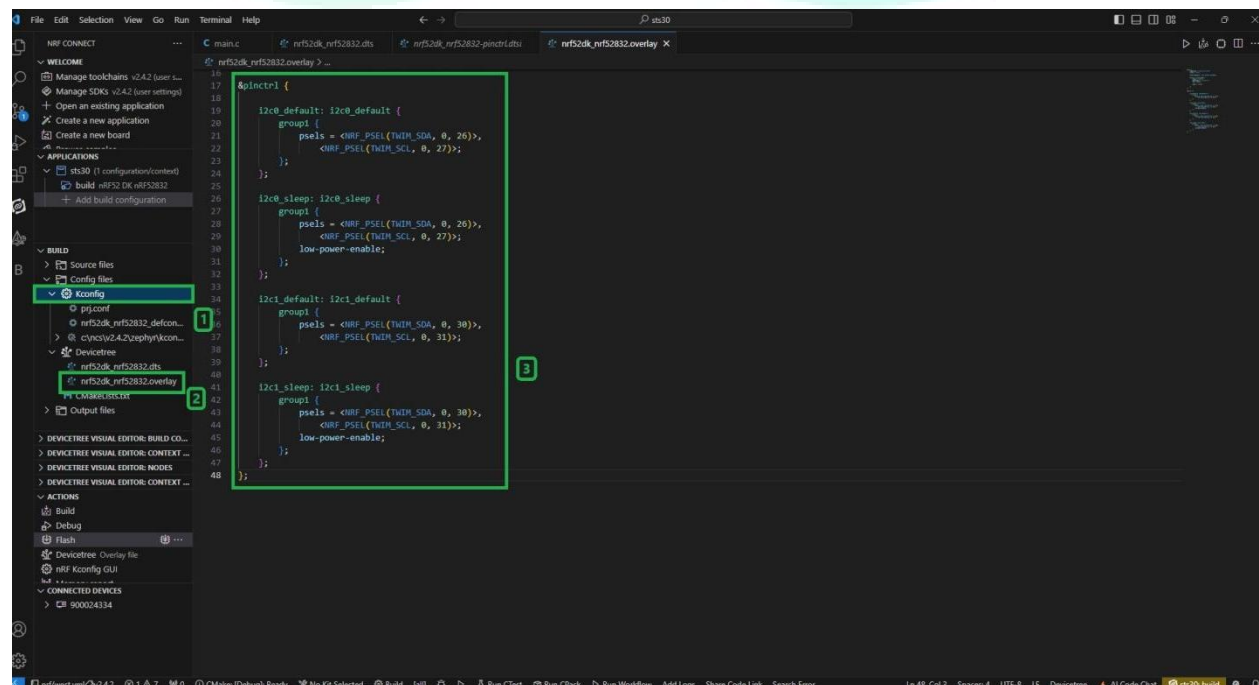
- Click on **Create new build configuration [1]**. Here you can change the board version, if you are using nRF52832, then select **nrf52dk_nrf52832 [2]** or you can change from dropdown menu for another version like nRF52833 etc.
- Click on the Configuration and select **prj.conf [3]** from dropdown menu and then click on the device tree overlays and select **nrf52dk_nrf52832.overlay[4]**.
- Then click on the **Build Configuration [5]** as shown below in the picture.



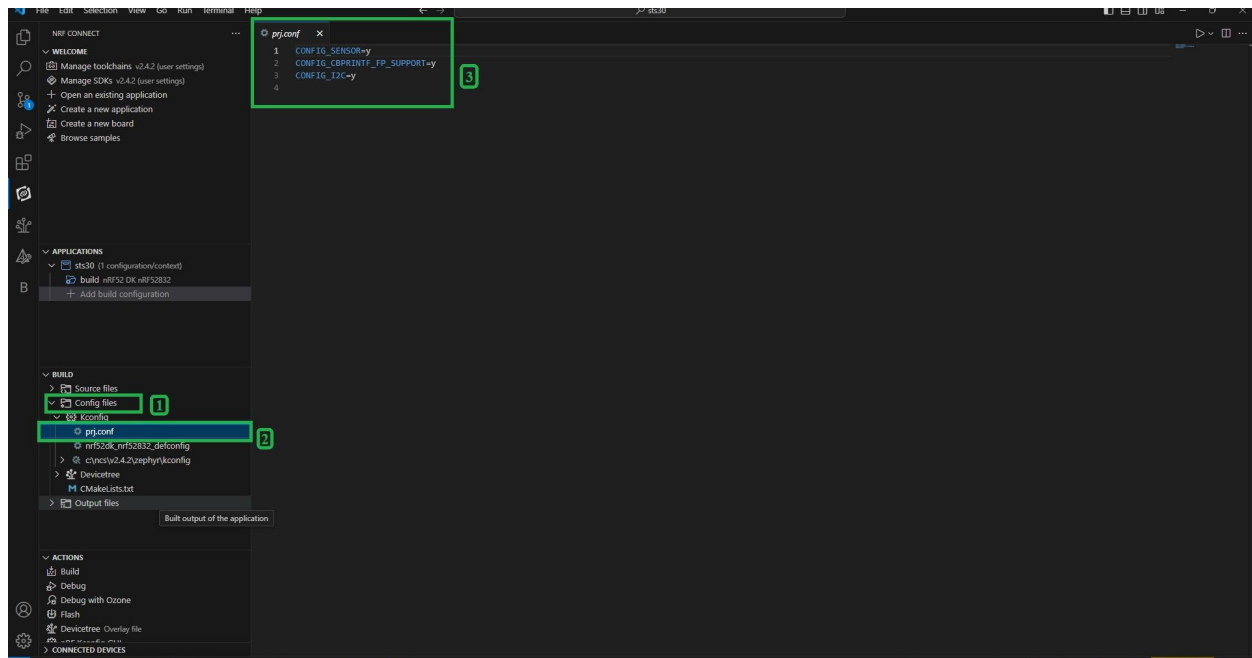
- Go to source file, click **source file [1]** > click on **Application** > click on **src** > click on **main.c [2]**.
- By clicking on **main.c** file and you will see the code on your screen.



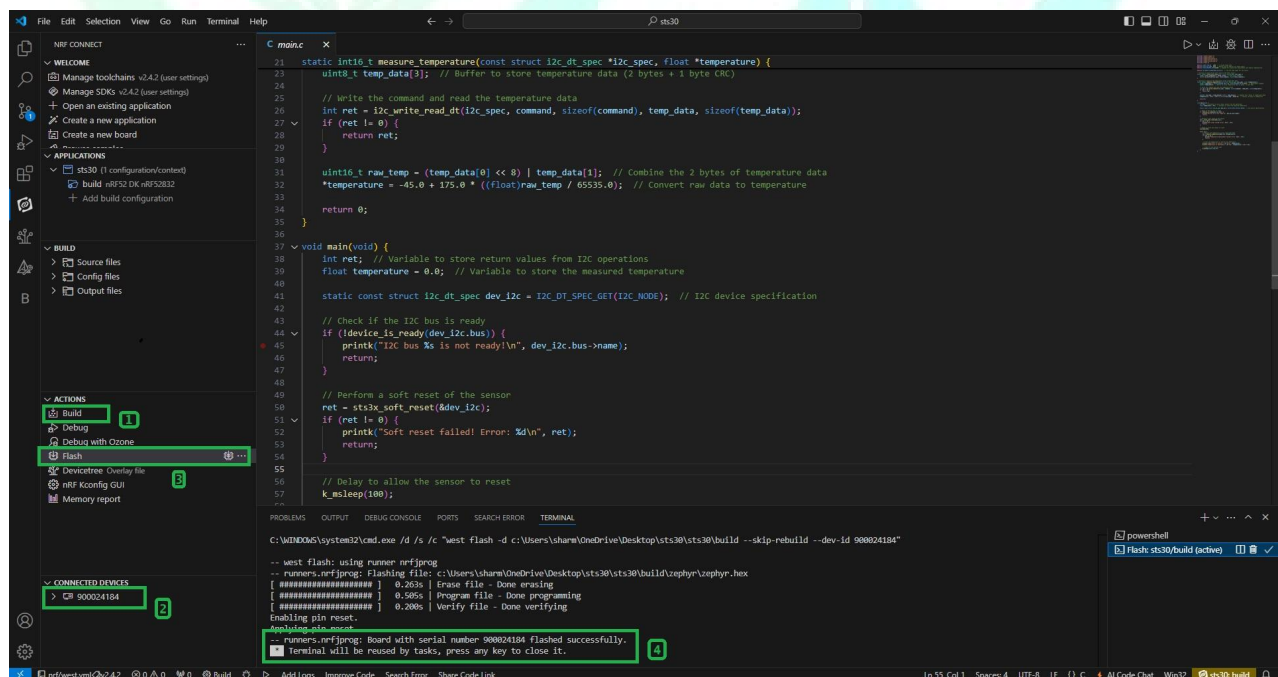
- To configure the i2c & UART protocols, you have to enable it in the **overlay file**.
- Click on the **Config files[1]** > click on **Kconfig** > click on **Devicetree** > click on **nrf52dk_nrf52832.overlay [2]**.
- The overlay file will appear on your screen and add the given code to the **overlay file** as shown in the picture given below [3].



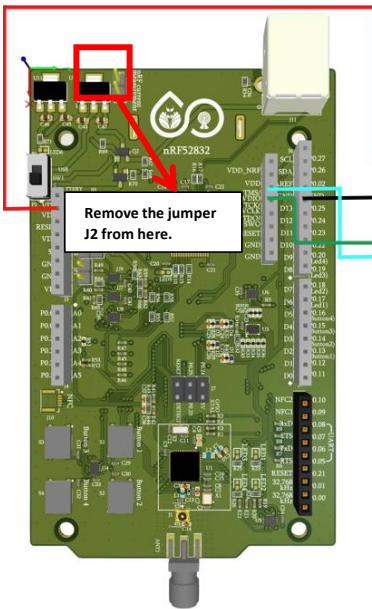
- You need to enable sensor in prj file for communication as shown below.
- Click on **Config files [1]** > then click on **Kconfig files** > click on **prj.conf [2]**.
- The **prj.conf** will appear on the screen [3] as shown below in the picture.



- Click on **Build [1]** configuration again and check the **CONNECTED DEVICES [2]**.
- If device id is visible, then **Flash [3]** the code in Dev Kit.



- For Node programming remove the jumper **J2** from the development board.
- Now flash the code with the help of nRF52832 development board as shown below in the figure.



Board Pins -> NODE Pins

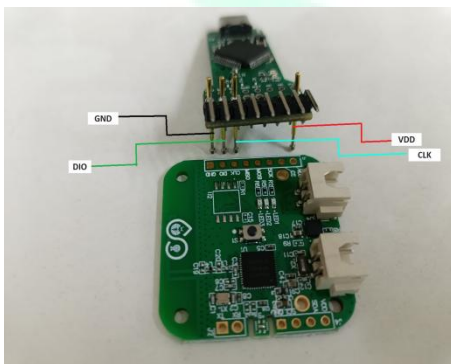
VDD(3.3V) -> VDD

GND -> GND

CLK -> CLK

DIO -> DIO

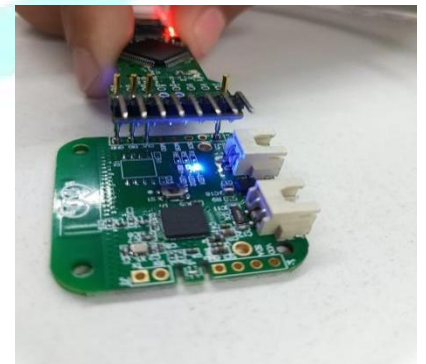
- There is another way of flashing the code with the help of Node Programmer as shown in the picture below.



- NODE without connection.

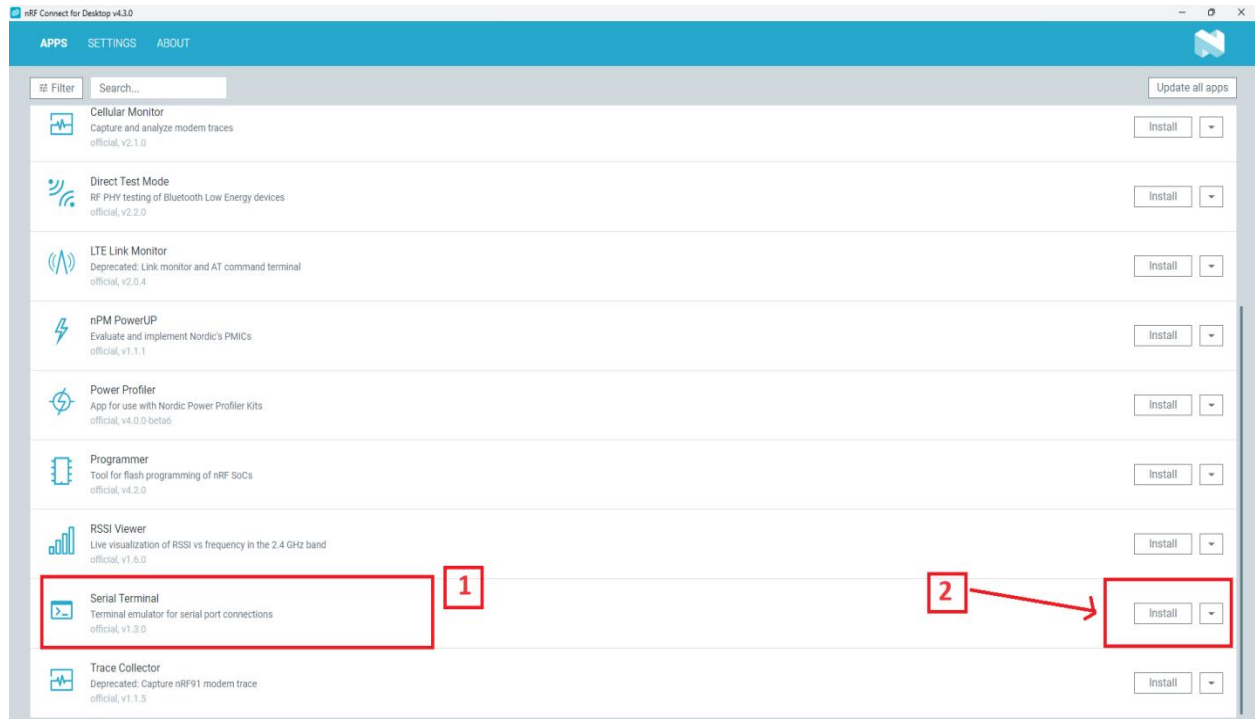


- NODE with connection.

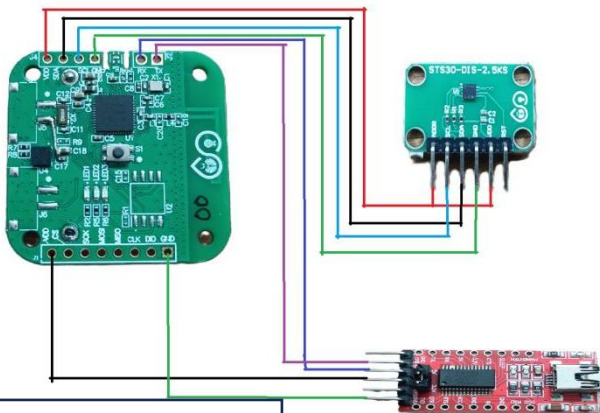


- NODE after program.

- Firstly, you have to **Install [2]** the nRF **Serial Terminal [1]** in nRF Connect for Desktop application as shown below.



- Connect the **TTL Device** for UART communication so that the data must appear on the serial terminal.
- Connect the **TTL Device** as shown below in the picture.



Note: - Do not supply 5 volts current to board through TTL. Otherwise, board will damage.

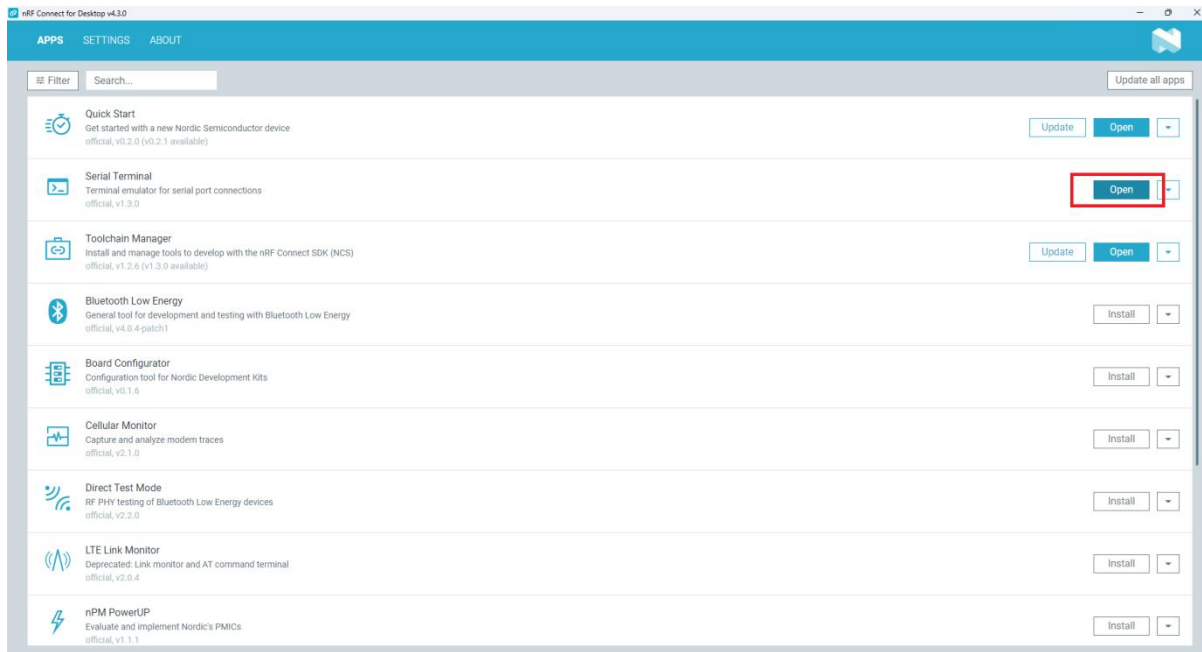
Node Pins -> TTL Pins

Tx -> Rx
Rx -> Tx
VDD -> VDD
GND -> GND

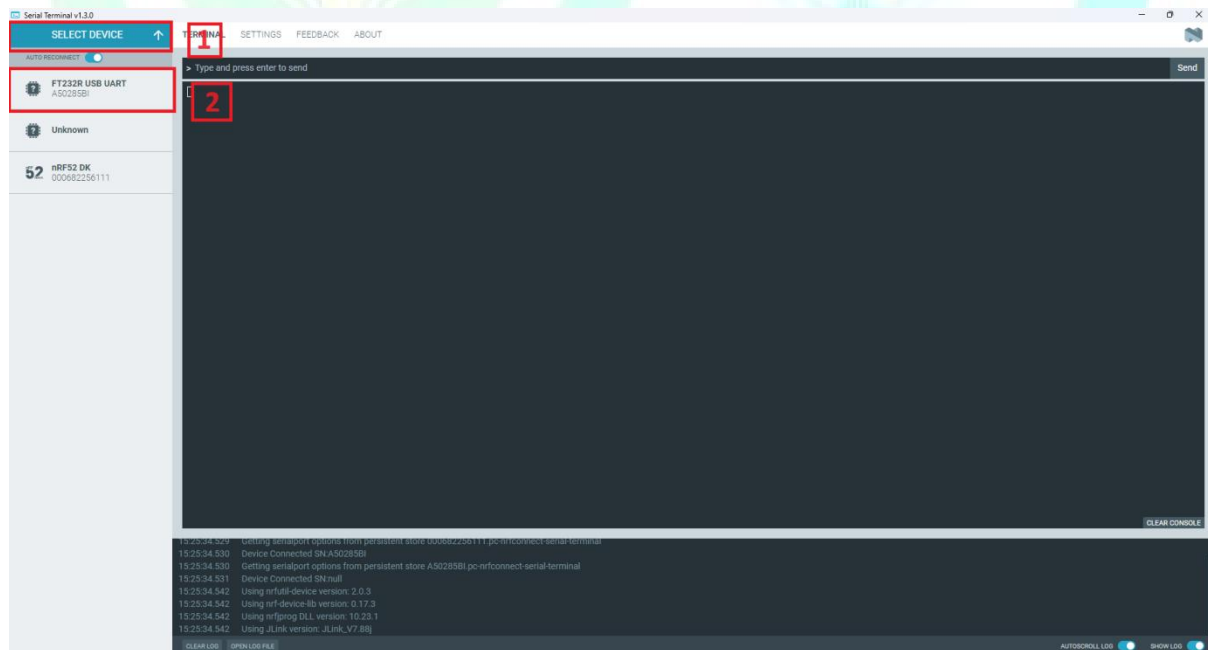
Node Pins -> Sensor Pins

SDA -> SDA
SCL -> SCL
VDD -> VDD
GND -> GND
VDD -> ADDR

- Click on **Open** as shown below in the picture.

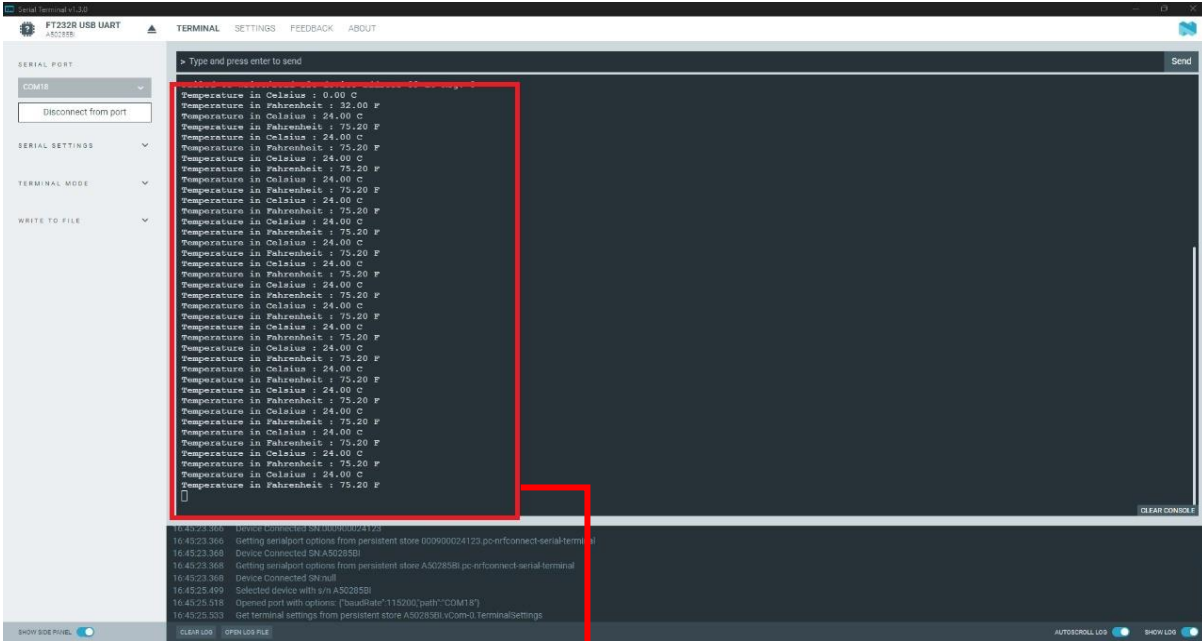


➤ Click on **Select Device** [1] > click on **FT232R USB UART** [2] as shown below in the picture.



➤ Now the output will appear on your screen as shown below.

OUTPUT

[illegible]